

# Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification — Community Lost & Found Matching Network (Problem Statement #59)

## Architecture Selection

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Microservices Architecture was chosen for the Community Lost & Found Matching Network System.

## Architectural Choice

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Microservices Architecture was selected, decomposed into independently deployable services: **Item Posting Service** and **Tag Extraction Service** (ingestion), **Matching Engine Service**, **Claim Verification Service** and **Moderation Service** (business logic), **Notification Service** (cross-cutting), and a **Listing Database Service** that backs every other service. Services are made to communicate only through the labelled interfaces on the component diagram (e.g. *ExtractTags*, *ListingQuery*, *MatchAlert*, *AnswerStore*); another service's data is never reached into directly.

## Two Reasons

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1. **Wildly uneven, independently scaled load.** Under NFR-002, the Matching Engine Service alone is required to sustain 500+ concurrent geo-proximity match requests within a 5-minute SLA, while Moderation and Notification are run at a far steadier, lower volume. If a Layered or Client-Server design were used, the whole application would be forced to be scaled as one unit; with microservices, only the Matching Engine (and, at upload time, the Tag Extraction Service, which is required to auto-tag photos within 10 seconds per FR-002) needs to be scaled out, without the rest of the system being over-provisioned.
2. **Independently evolvable capabilities.** Tag extraction (vision/NLP), geo-matching, claim verification and admin moderation are functionally unrelated and are changed on different schedules — the matching algorithm may be retrained often, while moderation rules are changed rarely. With microservices, each capability can be built, deployed and rolled back on its own release cycle, and a failure in one (e.g. Tag Extraction) can be degraded gracefully instead of posting, matching or claims being taken down.

## Security Advantage

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By isolating the **Claim Verification Service** as its own service, reachable only through its narrow *SubmitClaim/AnswerStore* interfaces, NFR-001 and FR-003 are directly supported: proof-of-ownership answers and the logic that reveals contact details are never held inside, and are never made directly reachable from, the Item Posting, Tag Extraction or Moderation services. If a lower-trust, high-churn service such as Tag Extraction is compromised, stored answers cannot be reached and a contact-detail reveal cannot be triggered — that path is confined to one service with one well-defined boundary, and the attack surface for the platform's most sensitive data is shrunk.

## Performance Benefit

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Because each service is deployed and scaled independently, the Matching Engine Service can be scaled out horizontally during posting-time traffic spikes so that the 95th-percentile match time is kept under 5 minutes at 500+ concurrent requests (NFR-002), without that extra capacity — or its cost — being forced onto the lighter-weight Notification or Moderation services. Through fault isolation, a match or a claim is never blocked from completing by a slow or failing Notification Service, so the core matching and claiming paths are kept responsive even when a downstream service is degraded.